Correction Factor: $CF = \frac{G^2}{n} = \frac{(66)^2}{12} = \frac{4356}{12} = 363$

Total Sum of Squares: $TSS = \sum\sum x_{ij}^2 - C.F$

$= 408 - 363 = 45$

Sum of Squares due to Treatments: $SST = \frac{\sum_{i=1}^{k} x_{.j}^2}{k} - C.F$

$= \frac{1140}{3} - 363$

$= 380 - 363 = 17$

Sum of Squares due to Blocks: $SSB = \frac{\sum_{i=1}^{k} x_{.j}^2}{k} - C.F$

$= \frac{1524}{4} - 363$

$= 381 - 363$

$= 18$

Sum of Squares due to Error: $SSE = TSS - SST - SSB$

$= 45 - 17 - 18 = 10$

Mean sum of squares: $MST = \frac{SST}{k-1} = \frac{17}{2} = 8.5$

$MSB = \frac{SSB}{m-1} = \frac{18}{3} = 6$

$MSE = \frac{SSE}{(k-1)(m-1)} = \frac{10}{6} = 1.67$

ANOVA Table (two-way)

Sources of variation	Sum of squares	Degrees of freedom	Mean sum of squares	*F*-ratio
Between treatments (Programs)	17	2	8.5	$F_{ot} = \frac{8.5}{1.67} = 5.09$
Between blocks (Salesmen)	18	3	6	$F_{ob} = \frac{6}{1.67} = 3.59$
Error	10	6	1.67	
Total		11		

Step 6 : Critical values

(i) $f_{(2, 6),0.05} = 5.1456$ (for treatments)

(ii) $f_{(3, 6),0.05} = 4.7571$ (for blocks)

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